Lab 12 practice: triple integrals and the divergence theorem (Zill §9.15, 9.16)

Prep for the Lab 12 quiz (week of Nov 30). Closed book, about 10 minutes per problem. Problems 1 and 2 are rectangular coordinates (planes, then surfaces), problem 3 is cylindrical, problem 4 is spherical, and problem 5 is about choosing the system. Problems 6 to 8 are the divergence theorem: verify both sides, close an open surface with a cap, and check the hypotheses. No figures are given, so sketch each solid before you set limits. State the coordinate system and the limits before you integrate. Every answer was recomputed by a different route (`lab12_check.py`).

Cylindrical. $x = r\cos\theta,\ y = r\sin\theta,\ z = z$, with $dV = r\,dz\,dr\,d\theta$ and $x^2 + y^2 = r^2$. Use it when the solid has an axis of symmetry along z: cylinders, paraboloids $z = x^2 + y^2$, discs as shadows.

Spherical. $x = \rho\sin\phi\cos\theta,\ y = \rho\sin\phi\sin\theta,\ z = \rho\cos\phi$, with $\rho \geq 0,\ 0 \leq \phi \leq \pi$, and $dV = \rho^2\sin\phi\,d\rho\,d\phi\,d\theta$ and $x^2 + y^2 + z^2 = \rho^2$. Use it for spheres, cones $z = c\sqrt{x^2 + y^2}$ (which is $\phi =$ constant) and anything that is a function of $x^2 + y^2 + z^2$.

Setting limits. Innermost variable first: its limits may use the other two. Each later limit may use only variables still to come. The outermost limits are constants.

Divergence theorem. For a solid D bounded by a closed, piecewise smooth surface S with outward unit normal $\mathbf{n}$, and $\mathbf{F}$ with continuous first partials on an open region containing D:

$$\iint_S \mathbf{F}\cdot\mathbf{n}\,dS = \iiint_D \nabla\cdot\mathbf{F}\,dV, \qquad \nabla\cdot\mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

Check all four hypotheses first: closed, piecewise smooth, outward orientation, $\mathbf{F}$ is C^1 everywhere inside D (including its interior points).

Problem 1

A solid tetrahedron D is bounded by the three coordinate planes and the plane $x + 2y + z = 4$. Its density is $\delta(x, y, z) = y$ kg/m^3. Set up the limits in the order $dz\,dy\,dx$, then find (a) the volume of D and (b) its mass.

Solution: rectangular coordinates, solid bounded by planes

Coordinates and limits. Rectangular. The plane meets the axes at $(4,0,0)$, $(0,2,0)$ and $(0,0,4)$, so D is the tetrahedron with those three vertices and the origin. For $dz\,dy\,dx$, z runs from the floor $z=0$ up to the plane, $z=4-x-2y$. The shadow in the xy-plane (set $z=0$) is the triangle $x\ge 0$, $y\ge 0$, $x+2y\le 4$, so

$$0\le x\le 4, \qquad 0\le y\le \frac{4-x}{2}, \qquad 0\le z\le 4-x-2y$$

Write $u=4-x$ to keep the algebra short, so the upper limit for y is $u/2$ and for z it is $u-2y$.

(a) Volume.

$$V=\int_0^4\int_0^{u/2}\int_0^{u-2y} dz\,dy\,dx=\int_0^4\int_0^{u/2}(u-2y)\,dy\,dx=\int_0^4\Big(\frac{u^2}{2}-\frac{u^2}{4}\Big)dx=\int_0^4\frac{(4-x)^2}{4}\,dx=\frac{64}{12}=\frac{16}{3}\ \mathrm{m}^3$$

This agrees with $\frac{1}{6}abc=\frac{1}{6}(4)(2)(4)$, the volume of a tetrahedron with perpendicular edges a,b,c.

(b) Mass.

$$m=\int_0^4\int_0^{u/2}\int_0^{u-2y} y\,dz\,dy\,dx=\int_0^4\int_0^{u/2}y(u-2y)\,dy\,dx=\int_0^4\Big[\frac{uy^2}{2}-\frac{2y^3}{3}\Big]_0^{u/2}dx$$

At $y=u/2$ the bracket is $\dfrac{u^3}{8}-\dfrac{u^3}{12}=\dfrac{u^3}{24}$, so

$$m=\int_0^4\frac{(4-x)^3}{24}\,dx=\frac{1}{24}\cdot\frac{4^4}{4}=\frac{8}{3}\ \mathrm{kg}$$

$$\boxed{V=\tfrac{16}{3}\ \mathrm{m}^3, \qquad m=\tfrac{8}{3}\ \mathrm{kg}}$$

Problem 2

Find the volume of the solid bounded by the parabolic cylinder $y = x^2$, the plane $y + z = 4$ and the plane $z = 0$. Give the limits in the order $dz\,dy\,dx$, then redo the setup in the order $dz\,dx\,dy$ and confirm the value.

Solution: rectangular coordinates, solid bounded by surfaces

Coordinates and limits. Rectangular. The solid sits over the region of the xy-plane trapped between the parabola $y = x^2$ and the line $y = 4$ (the plane $y + z = 4$ has $z = 0$ there). The parabola meets $y = 4$ at $x = \pm 2$. Height: from $z = 0$ up to $z = 4 - y$.

Order $dz\,dy\,dx$:

$$-2 \le x \le 2, \qquad x^2 \le y \le 4, \qquad 0 \le z \le 4 - y$$

$$V = \int_{-2}^{2} \int_{x^2}^{4} (4 - y)\, dy\, dx = \int_{-2}^{2} \frac{(4 - x^2)^2}{2}\, dx = \frac{1}{2} \int_{-2}^{2} (16 - 8x^2 + x^4) dx$$

$$= \frac{1}{2}\Big[16x - \frac{8x^3}{3} + \frac{x^5}{5}\Big]_{-2}^{2} = \frac{1}{2}\Big(64 - \frac{128}{3} + \frac{64}{5}\Big) = \frac{256}{15}$$

Other order, $dz\,dx\,dy$. Now y is outermost and runs over its full range, $0 \le y \le 4$. For a fixed y, x runs across the parabola: $-\sqrt{y} \le x \le \sqrt{y}$. Then $0 \le z \le 4 - y$.

$$V = \int_0^4 \int_{-\sqrt{y}}^{\sqrt{y}} (4 - y)\, dx\, dy = \int_0^4 2\sqrt{y}\,(4 - y)\, dy = \int_0^4 (8y^{1/2} - 2y^{3/2}) dy = \Big[\frac{16}{3} y^{3/2} - \frac{4}{5} y^{5/2}\Big]_0^4 = \frac{128}{3} - \frac{128}{5} = \frac{256}{15}$$

$$\boxed{V = \tfrac{256}{15} \text{ units}^3 \approx 17.07}$$

Problem 3

The solid D lies between the paraboloids $z = x^2 + y^2$ (below) and $z = 8 - x^2 - y^2$ (above). Find (a) its volume and (b) its mass if the density is $\delta = z$.

Solution: cylindrical coordinates, paraboloids

Coordinates and limits. Cylindrical, because both surfaces are functions of $r^2 = x^2 + y^2$. The paraboloids meet where $r^2 = 8 - r^2$, so $r = 2$ (and $z = 4$). The shadow in the xy-plane is the disc $r \le 2$. For r in that disc the lower surface is $z = r^2$ and the upper is $z = 8 - r^2$:

$$0 \le \theta \le 2\pi, \qquad 0 \le r \le 2, \qquad r^2 \le z \le 8 - r^2, \qquad dV = r\,dz\,dr\,d\theta$$

(a) Volume.

$$V = \int_0^{2\pi} \int_0^2 \int_{r^2}^{8-r^2} r\,dz\,dr\,d\theta = 2\pi \int_0^2 r\,(8 - 2r^2)\,dr = 2\pi \Big[4r^2 - \frac{r^4}{2}\Big]_0^2 = 2\pi(16 - 8) = 16\pi$$

(b) Mass. Integrate $z\,r$ over the same limits.

$$m = \int_0^{2\pi} \int_0^2 \int_{r^2}^{8-r^2} z\,r\,dz\,dr\,d\theta = 2\pi \int_0^2 r \cdot \frac{(8 - r^2)^2 - r^4}{2}\,dr$$

The numerator is $(8 - r^2)^2 - r^4 = 64 - 16r^2$, so

$$m = 2\pi \int_0^2 r\,(32 - 8r^2)\,dr = 2\pi \Big[16r^2 - 2r^4\Big]_0^2 = 2\pi(64 - 32) = 64\pi$$

$$\boxed{V = 16\pi, \qquad m = 64\pi}$$

Problem 4

Let D be the ball $x^2 + y^2 + z^2 \leq 4z$. Find (a) its volume and (b) $\iiint_D z\, dV$, then use them to get the height of its centroid.

Solution: spherical coordinates, sphere not centred at the origin

Coordinates and limits. Spherical. Convert the boundary: $\rho^2 = 4\rho\cos\phi$, so $\rho = 4\cos\phi$. (Completing the square gives $x^2 + y^2 + (z-2)^2 = 4$, a ball of radius 2 centred at $(0,0,2)$, which touches the origin.) The ball lies in $z \geq 0$, so ϕ only reaches $\pi/2$ and $\cos\phi \geq 0$:

$$0 \leq \theta \leq 2\pi, \qquad 0 \leq \phi \leq \frac{\pi}{2}, \qquad 0 \leq \rho \leq 4\cos\phi, \qquad dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

(a) Volume.

$$V = \int_0^{2\pi} \int_0^{\pi/2} \int_0^{4\cos\phi} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = 2\pi \int_0^{\pi/2} \frac{64\cos^3\phi}{3} \sin\phi \, d\phi = 2\pi \cdot \frac{64}{3} \cdot \frac{1}{4} = \frac{32\pi}{3}$$

This matches $\frac{4}{3}\pi(2)^3$.

(b) The integral of z. With $z = \rho\cos\phi$ the integrand is $\rho^3 \cos\phi \sin\phi$:

$$\iiint_D z \, dV = 2\pi \int_0^{\pi/2} \cos\phi \sin\phi \, \frac{(4\cos\phi)^4}{4} \, d\phi = 2\pi \cdot 64 \int_0^{\pi/2} \cos^5\phi \sin\phi \, d\phi = 2\pi \cdot 64 \cdot \frac{1}{6} = \frac{64\pi}{3}$$

Centroid height. $\bar{z} = \dfrac{64\pi/3}{32\pi/3} = 2$, the centre of the ball, as symmetry says.

$$\boxed{V = \tfrac{32\pi}{3}, \qquad \iiint_D z \, dV = \tfrac{64\pi}{3}, \qquad \bar{z} = 2}$$

Problem 5

Evaluate

$$\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{8-x^2-y^2}} z \, dz \, dy \, dx$$

Describe the solid in words, choose a coordinate system and say why, then confirm with the other curvilinear system.

Solution: reading the region, then choosing the system

The solid. In z the region runs from the cone $z = \sqrt{x^2 + y^2}$ up to the sphere $x^2 + y^2 + z^2 = 8$ (upper half, radius $2\sqrt{2}$). The cone and sphere meet where $r^2 + r^2 = 8$, so $r = 2$, matching the shadow disc $x^2 + y^2 \le 4$ in the outer two limits. So D is an ice-cream cone: the part of the ball of radius $2\sqrt{2}$ lying above the 45° cone.

Choice. Spherical is better: the cone is $\phi = \pi/4$ (since $z = r$ means $\tan\phi = 1$) and the sphere is $\rho = 2\sqrt{2}$, so every limit is a constant. In Cartesian form the integral is nearly impossible to do by hand.

Spherical.

$$0 \le \theta \le 2\pi, \qquad 0 \le \phi \le \frac{\pi}{4}, \qquad 0 \le \rho \le 2\sqrt{2}$$

$$\iiint_D z\,dV = \int_0^{2\pi} \int_0^{\pi/4} \int_0^{2\sqrt{2}} \rho\cos\phi\ \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$$

The integral separates into three single integrals:

$$= 2\pi \cdot \Big[\frac{\rho^4}{4}\Big]_0^{2\sqrt{2}} \cdot \Big[\frac{\sin^2\phi}{2}\Big]_0^{\pi/4} = 2\pi \cdot 16 \cdot \frac{1}{4} = 8\pi$$

Cylindrical, as a check. $0 \le \theta \le 2\pi$, $0 \le r \le 2$, $r \le z \le \sqrt{8 - r^2}$:

$$2\pi \int_0^2 r \cdot \frac{(8 - r^2) - r^2}{2}\,dr = 2\pi \int_0^2 (4r - r^3)dr = 2\pi(8 - 4) = 8\pi$$

$$\boxed{\iiint_D z\,dV = 8\pi}$$

Cylindrical works too, because the cone becomes $z = r$, but its upper limit $\sqrt{8 - r^2}$ is messier than the constant $2\sqrt{2}$.

Problem 6

Let $\mathbf{F} = \langle x^3,\ y^3,\ z^2 \rangle$ and let S be the closed surface made of the cylinder $x^2 + y^2 = 4$ for $0 \leq z \leq 3$, together with the discs $z = 0$ and $z = 3$ that cap it. Verify the divergence theorem by computing both sides, with S oriented outward.

Solution: verifying both sides

Hypotheses. S is closed (the two discs seal the cylinder), piecewise smooth (three smooth pieces), and given the outward orientation. $\mathbf{F}$ is polynomial, so it is C^1 everywhere. The theorem applies.

Volume side. $\nabla \cdot \mathbf{F} = 3x^2 + 3y^2 + 2z$. Cylindrical coordinates, since D is a solid cylinder about the z-axis:

$$0 \le \theta \le 2\pi, \qquad 0 \le r \le 2, \qquad 0 \le z \le 3, \qquad \nabla \cdot \mathbf{F} = 3r^2 + 2z$$

$$\iiint_D \nabla \cdot \mathbf{F}\, dV = \int_0^{2\pi} \int_0^2 \int_0^3 (3r^2 + 2z)\, r\, dz\, dr\, d\theta = 2\pi \int_0^2 (9r^3 + 9r)\, dr = 2\pi(36 + 18) = 108\pi$$

Surface side. Three pieces, each with its outward normal.

Lateral surface. $\mathbf{n} = \langle x, y, 0\rangle/2$. On it $x = 2\cos\theta$, $y = 2\sin\theta$, and $dS = 2\, d\theta\, dz$ (arc length is $r\, d\theta = 2\, d\theta$):

$$\mathbf{F} \cdot \mathbf{n} = \frac{x^4 + y^4}{2} = 8(\cos^4\theta + \sin^4\theta)$$

$$\iint_{\text{lat}} = \int_0^3 \int_0^{2\pi} 16(\cos^4\theta + \sin^4\theta)\, d\theta\, dz = 3 \cdot 16 \cdot \Big(\frac{3\pi}{4} + \frac{3\pi}{4}\Big) = 72\pi$$

using $\int_0^{2\pi} \cos^4\theta\, d\theta = \int_0^{2\pi} \sin^4\theta\, d\theta = \frac{3\pi}{4}$.

Top disc, $z = 3$. $\mathbf{n} = \mathbf{k}$, $\mathbf{F} \cdot \mathbf{n} = z^2 = 9$, area 4π: flux $= 36\pi$.

Bottom disc, $z = 0$. $\mathbf{n} = -\mathbf{k}$, $\mathbf{F} \cdot \mathbf{n} = -z^2 = 0$: flux $= 0$.

$$72\pi + 36\pi + 0 = 108\pi$$

Both sides agree.

$$\boxed{\iint_S \mathbf{F} \cdot \mathbf{n}\, dS = \iiint_D \nabla \cdot \mathbf{F}\, dV = 108\pi}$$

Problem 7

Let $\mathbf{F} = \langle xz,\ yz,\ z^2+1 \rangle$ and let S be the paraboloid $z = 4-x^2-y^2$ with $z \geq 0$, oriented upward (away from the z-axis, outward from the solid under it). Find $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$ by closing S with a cap, applying the divergence theorem to the closed surface, and subtracting the flux through the cap.

Solution: closing an open surface

Why it needs a cap. S is a bowl turned upside down with nothing across the bottom, so it is not closed and the theorem does not apply to it directly. Add the disc C: $z = 0$, $x^2 + y^2 \le 4$. Then $S \cup C$ is closed, piecewise smooth, outward-oriented (upward on S, downward on C), and $\mathbf{F}$ is polynomial, hence C^1. All hypotheses hold for the solid D under the paraboloid, and

$$\iint_S + \iint_C = \iiint_D \nabla \cdot \mathbf{F}\, dV \qquad \text{so} \qquad \iint_S = \iiint_D \nabla \cdot \mathbf{F}\, dV - \iint_C$$

Volume side. $\nabla \cdot \mathbf{F} = z + z + 2z = 4z$. Cylindrical coordinates, since D is bounded by a paraboloid about the z-axis and a disc:

$$0 \le \theta \le 2\pi, \qquad 0 \le r \le 2, \qquad 0 \le z \le 4 - r^2$$

$$\iiint_D 4z\, dV = 2\pi \int_0^2 r \left[2z^2\right]_0^{4-r^2} dr = 4\pi \int_0^2 (4 - r^2)^2\, r\, dr$$

Put $u = 4 - r^2$, so $r\, dr = -\frac{1}{2} du$, and u runs from 4 down to 0:

$$= 4\pi \cdot \frac{1}{2} \int_0^4 u^2\, du = 2\pi \cdot \frac{64}{3} = \frac{128\pi}{3}$$

Cap flux. On C, $\mathbf{n} = -\mathbf{k}$ and $z = 0$, so $\mathbf{F} \cdot \mathbf{n} = -(z^2 + 1) = -1$. The disc has area 4π, so $\iint_C = -4\pi$.

Subtract.

$$\iint_S \mathbf{F} \cdot \mathbf{n}\, dS = \frac{128\pi}{3} - (-4\pi) = \frac{140\pi}{3}$$

Check by direct computation. For $z = g(x, y) = 4 - x^2 - y^2$, the upward normal has $\mathbf{n}\, dS = \langle -g_x, -g_y, 1\rangle\, dA = \langle 2x, 2y, 1\rangle\, dA$. Then $\mathbf{F} \cdot \mathbf{n}\, dS = (2x^2 z + 2y^2 z + z^2 + 1) dA = (2r^2(4 - r^2) + (4 - r^2)^2 + 1) dA = (17 - r^4)\, dA$, and

$$2\pi \int_0^2 (17 - r^4)\, r\, dr = 2\pi\Big(34 - \frac{32}{3}\Big) = \frac{140\pi}{3}$$

$$\boxed{\iint_S \mathbf{F} \cdot \mathbf{n}\, dS = \tfrac{140\pi}{3}}$$

Problem 8

Let $\mathbf{F} = \dfrac{\langle x, y, z\rangle}{(x^2+y^2+z^2)^{3/2}}$. (a) Show $\nabla \cdot \mathbf{F} = 0$ wherever $\mathbf{F}$ is defined. (b) A student applies the divergence theorem to the sphere $x^2 + y^2 + z^2 = 4$ and concludes the outward flux is 0. Check the hypotheses, and compute the flux directly. (c) Does the theorem apply to the closed surface bounding the cube $1 \le x, y, z \le 2$? What does it give?

Solution: checking hypotheses

(a) Divergence. Write $\rho^2 = x^2 + y^2 + z^2$, so $\mathbf{F} = \langle x, y, z\rangle \rho^{-3}$ and $\partial\rho/\partial x = x/\rho$. Then

$$\frac{\partial}{\partial x}(x\rho^{-3}) = \rho^{-3} - 3x\rho^{-4} \cdot \frac{x}{\rho} = \rho^{-3} - \frac{3x^2}{\rho^5}$$

The y and z terms are the same with y^2, z^2. Add them:

$$\nabla \cdot \mathbf{F} = \frac{3}{\rho^3} - \frac{3(x^2+y^2+z^2)}{\rho^5} = \frac{3}{\rho^3} - \frac{3}{\rho^3} = 0 \qquad (\rho \neq 0)$$

(b) The sphere. Hypotheses: the sphere is closed, smooth, and the orientation is outward. All fine. The last one fails: $\mathbf{F}$ is not even defined at the origin, which is an interior point of the ball, so $\mathbf{F}$ is not C^1 on the solid. The theorem does not apply, and the student's 0 comes from integrating $\nabla \cdot \mathbf{F} = 0$ over a region where the divergence does not exist at one point. Compute the flux directly instead. On the sphere $\rho = 2$, $\mathbf{n} = \langle x, y, z\rangle/\rho$ and

$$\mathbf{F} \cdot \mathbf{n} = \frac{x^2+y^2+z^2}{\rho^3 \cdot \rho} = \frac{1}{\rho^2} = \frac{1}{4}$$

The area of the sphere is $4\pi(2)^2 = 16\pi$, so

$$\iint_S \mathbf{F} \cdot \mathbf{n}\, dS = \frac{1}{4} \cdot 16\pi = 4\pi \neq 0$$

The same value 4π comes out for a sphere of any radius R, since $\mathbf{F} \cdot \mathbf{n} = 1/R^2$ and the area is $4\pi R^2$.

(c) The cube. The solid $1 \leq x, y, z \leq 2$ does not contain the origin, so $\mathbf{F}$ is C^1 on an open region around it. The boundary is closed (six square faces), piecewise smooth, and can be oriented outward. All hypotheses hold, and

$$\iint_{\partial\text{cube}} \mathbf{F} \cdot \mathbf{n}\, dS = \iiint_{\text{cube}} 0\, dV = 0$$

Here the incoming flux through the three faces nearest the origin cancels the outgoing flux through the other three. The sphere in (b) and the cube in (c) differ in exactly one respect, whether the singular point lies inside.

$\boxed{\nabla \cdot \mathbf{F} = 0; \quad \text{sphere: theorem does not apply, flux} = 4\pi; \quad \text{cube: theorem applies, flux} = 0}$